

ALAN THOMAS

📍 Kerala, India ✉ alanthomask77@gmail.com ☎ +91 8281132762 🌐 in/alan-thomas-b30885276 🖱 alan-thomas-portfolio.vercel.app/

SUMMARY

Passionate and results-driven **Software Developer** with hands-on experience in **Cloud Computing**, **Artificial Intelligence**, **Machine Learning**, **Full Stack Development**, and **Python Development** through academic projects, cloud certifications, and practical implementations. Experienced in developing intelligent applications using **Python**, **TensorFlow**, **Keras**, **React.js**, **Node.js**, **SQL**, **MongoDB**, and **REST APIs**, while applying **Object-Oriented Programming (OOP)**, **Data Structures**, **Algorithms**, **Software Development Life Cycle (SDLC)**, and **Problem Solving** principles to deliver scalable software solutions.

Skilled in building **Deep Learning** models, **Time Series Forecasting**, **Computer Vision** applications, and cloud-ready software using **Amazon Web Services (AWS)**, **Oracle Cloud Infrastructure (OCI)**, **Google Cloud Platform (GCP)**, **Azure**, **Git**, **GitHub**, **Linux**, and **Prompt Engineering**. Passionate about designing cloud-native applications, AI-powered solutions, and modern software systems while continuously expanding expertise in cloud technologies, automation, and intelligent software development.

SKILLS

Software Development, Object-Oriented Programming (OOP), Data Structures, Algorithms, Software Development Life Cycle (SDLC), Problem Solving, Application Development, Debugging & Troubleshooting, Code Optimization

Artificial Intelligence & Machine Learning

Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Convolutional Neural Networks (CNN), Time Series Forecasting, Sentiment Analysis, Feature Engineering, Model Evaluation, Predictive Analytics

Cloud Computing

Cloud Computing, Cloud Infrastructure, Cloud Deployment, Cloud Security Fundamentals, Cloud-Native Development, Infrastructure Management, Cloud Support, Virtualization

Full Stack Development

Frontend Development, Backend Development, React Development, REST API Development, Responsive Web Development, Application Integration, User Interface Development

Data Science

Data Preprocessing, Data Cleaning, Data Visualization, Statistical Analysis, Exploratory Data Analysis (EDA), Data Modeling, Business Analytics

Professional Skills

Analytical Thinking, Communication, Problem Solving, Critical Thinking, Team Collaboration, Adaptability, Time Management, Continuous Learning

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript, SQL

AI & Machine Learning: TensorFlow, Keras, Scikit-learn, CNN, LSTM, Random Forest, Deep Learning, Machine Learning, Feature Engineering

Data Science: Pandas, NumPy, Matplotlib

Frontend Technologies: React.js, HTML5, CSS3, Tailwind CSS

Backend Technologies: Node.js, REST APIs

Cloud Platforms: Amazon Web Services (AWS), Oracle Cloud Infrastructure (OCI), Google Cloud Platform (GCP), Microsoft Azure

Databases: MySQL, MongoDB, SQL

Development Tools: Git, GitHub, Visual Studio Code (VS Code), Jupyter Notebook, Streamlit, Linux

AI & Productivity Tools: Prompt Engineering, AI Productivity Tools

PROJECT

Bitcoin Price Prediction Using Machine Learning

Python | LSTM | Random Forest | Pandas | NumPy | Matplotlib

- **Developed** a hybrid **Machine Learning** model combining **LSTM Deep Learning** and **Random Forest** algorithms to forecast Bitcoin price trends.
- **Integrated Twitter Sentiment Analysis** to improve prediction accuracy by incorporating real-time market sentiment.
- **Performed** comprehensive **Data Preprocessing**, feature engineering, normalization, and time-series transformation using **Pandas** and **NumPy**.
- **Evaluated** prediction accuracy using **RMSE** and **MAE** metrics while optimizing model performance.
- **Visualized** historical and predicted market trends using **Matplotlib**, enabling effective financial data interpretation.
- **Applied** predictive analytics techniques to improve forecasting reliability and support investment decision-making.
- **Documented** project architecture, machine learning workflow, and evaluation methodology for future enhancements.

Handwritten Digit Recognition Using CNN

Python | TensorFlow | Keras | CNN | Streamlit

- **Developed** a **Convolutional Neural Network (CNN)** model using **TensorFlow** and **Keras** to classify handwritten digits.

- **Trained** the model on the **MNIST Dataset** containing over **60,000 training images**, achieving high classification accuracy.
- **Performed** image normalization, tensor reshaping, and one-hot encoding to optimize model training.
- **Built** a multi-layer deep learning architecture using **Conv2D, MaxPooling, Dropout, and Dense Layers**.
- **Deployed** the trained model as an interactive **Streamlit Web Application** enabling real-time handwritten digit recognition.
- **Optimized** model performance through iterative evaluation and hyperparameter tuning.
- **Documented** implementation methodology, deep learning architecture, and deployment workflow.

Personal Portfolio Website

React.js | JavaScript | HTML5 | CSS3

- **Designed** and **developed** a fully responsive personal portfolio website using **React.js, JavaScript, HTML5, and CSS3**.
- **Built** modern user interfaces showcasing projects, technical skills, certifications, education, and professional achievements.
- **Integrated** GitHub repositories, project demonstrations, downloadable resume, and contact functionality.
- **Optimized** website responsiveness for desktop, tablet, and mobile devices.
- **Deployed** the application on **Vercel**, ensuring high availability and fast loading performance.
- **Applied** responsive web design principles to enhance user experience.
- **Maintained** and continuously improved the portfolio through version-controlled development workflows.

EDUCATION

Master of Computer Applications (MCA)-Pursuing

University of Calicut • Kerala, India

Bachelor of Computer Applications (BCA)

Bharathiar University • Kerala, India • 2025

Diploma in Cyber Security

JB International Skill Park • Kerala, India • 2025

Diploma in Cloud Computing

JB International Skill Park • Kerala, India • 2024

CERTIFICATIONS

Oracle Cloud Infrastructure (OCI) 2023 Certified Foundations Associate – Oracle

Getting Started with Artificial Intelligence – IBM SkillsBuild

Welcome to ServiceNow – ServiceNow

One Million Prompters Initiative – Certificate of Completion

Robotic Process Automation (RPA) Foundation Course – Naan Mudhalvan

Understanding and Securing Your Digital Identity – TCS iON Industry Project
